

Black Bear and Newfoundland Dog Image Recognition

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ACM40960 Projects in Maths Modelling, MSc Data and Computational Science



MOTIVATION

The project aims to distinguish similar images using advanced deep learning techniques, specifically leveraging **the DenseNet121 architecture** for high-accuracy image classification. The benefits include the following:

- **Wildlife Monitoring:** Improved accuracy for tracking and studying wildlife.
- **Safety:** Enhanced safety measures through precise species identification.
- **Conservation Efforts:** Superior data for developing conservation strategies.

DATASET

The dataset used in this project consists of 215 images of black bears and 371 images of Newfoundland dogs. The images are collected from public image databases, including different angles, lighting conditions, and backgrounds. Stratified sampling is used when dividing the subsets. They are divided into three subsets:

Training Set(64%): 374 images

Validation Set(16%): 94 images

Test Set(20%): 118 images

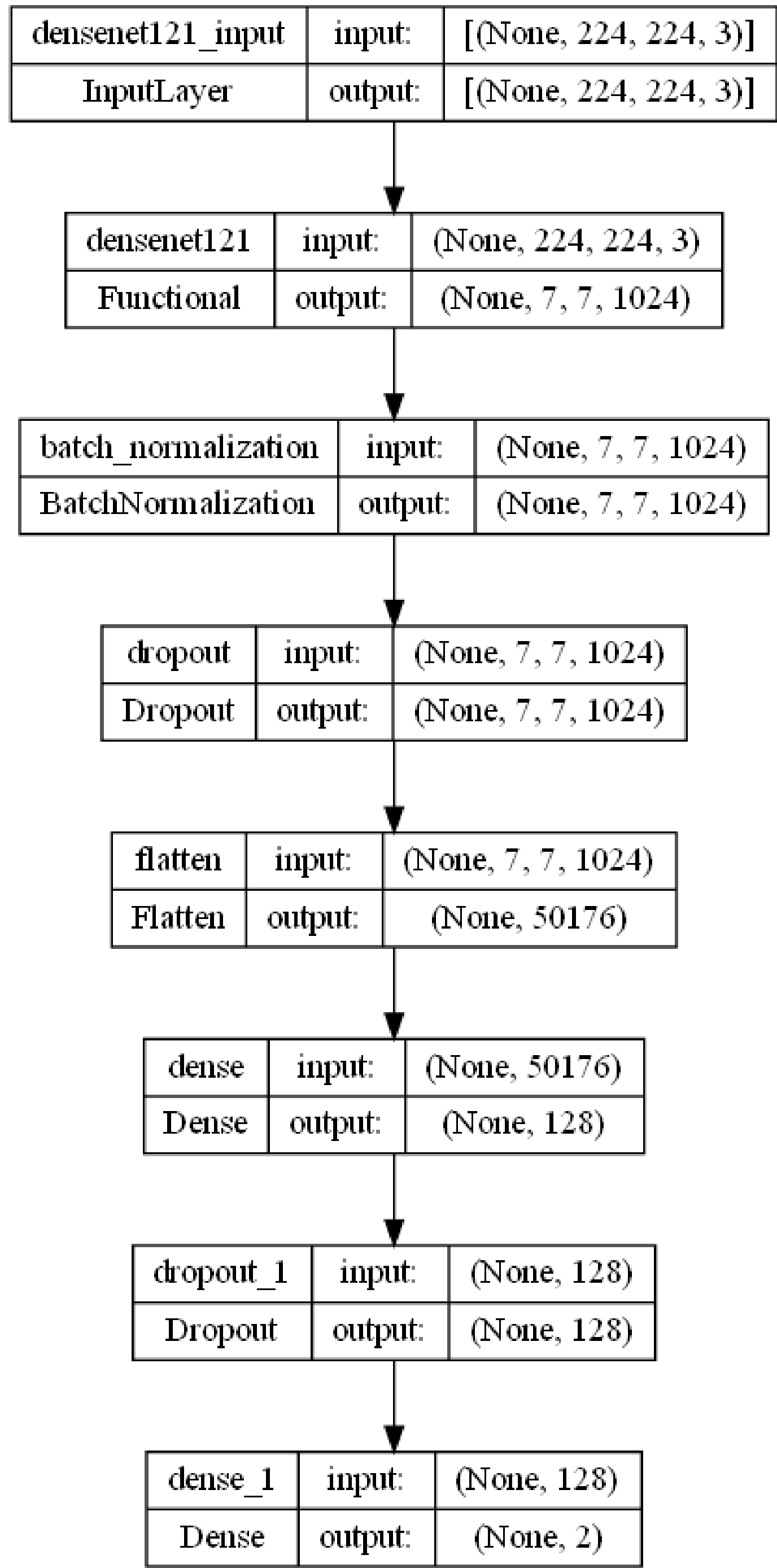


Figura 1: Black Bear



Figura 2: Newfoundland Dog

MODEL



Model Description:

Input Layer:

Accepts images of size 224x224x3.

DenseNet121 Base:

The pre-trained DenseNet121 network.

Batch Normalization:

Normalizes the output of the DenseNet121 network.

Dropout:

Prevents overfitting by randomly dropping neurons during training.

Flatten Layer:

Flattens the 2D feature maps into a 1D feature vector.

Dense Layers:

Fully connected layers with ReLU activation functions.

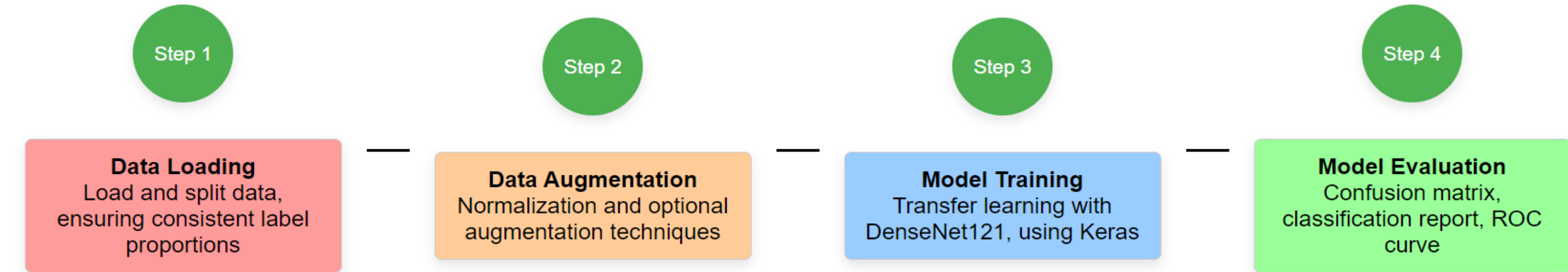
Dropout:

Prevents overfitting by randomly dropping neurons during training.

Output Layer:

Produces probabilities for the two classes using softmax activation.

METHODOLOGY



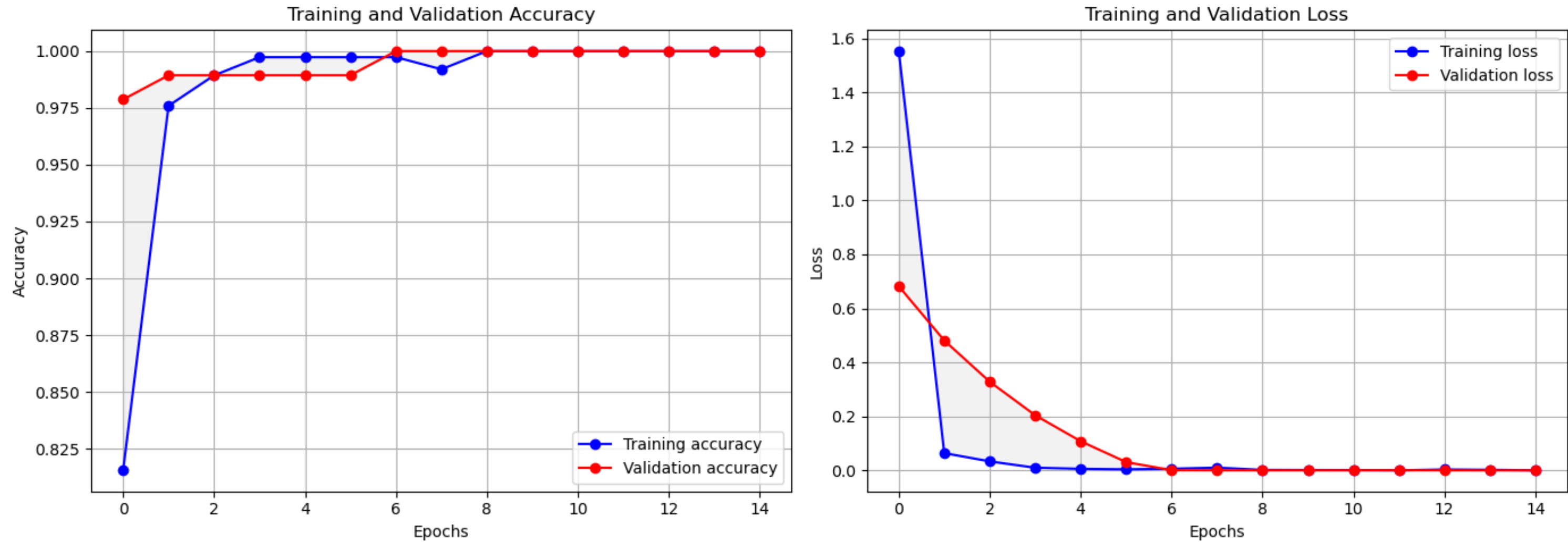
Model Highlights

- ◆ Stratified sampling ensures a consistent proportion of labels.
- ◆ All images are scaled and normalized.
- ◆ Using DenseNet121 as the base model, adding BatchNormalization, Dropout, and Dense layers, and employing the EarlyStopping callback function to monitor validation loss.
- ◆ Using various visualization methods to show the model's performance.

EVALUATION AND VISUALIZATION

Training & Validation Loss and Accuracy

The training and validation accuracy increase steadily and plateau around epoch 8, reaching nearly 100%. The training and validation loss decrease significantly and stabilize after epoch 6, indicating a well-fitting model with minimal overfitting.



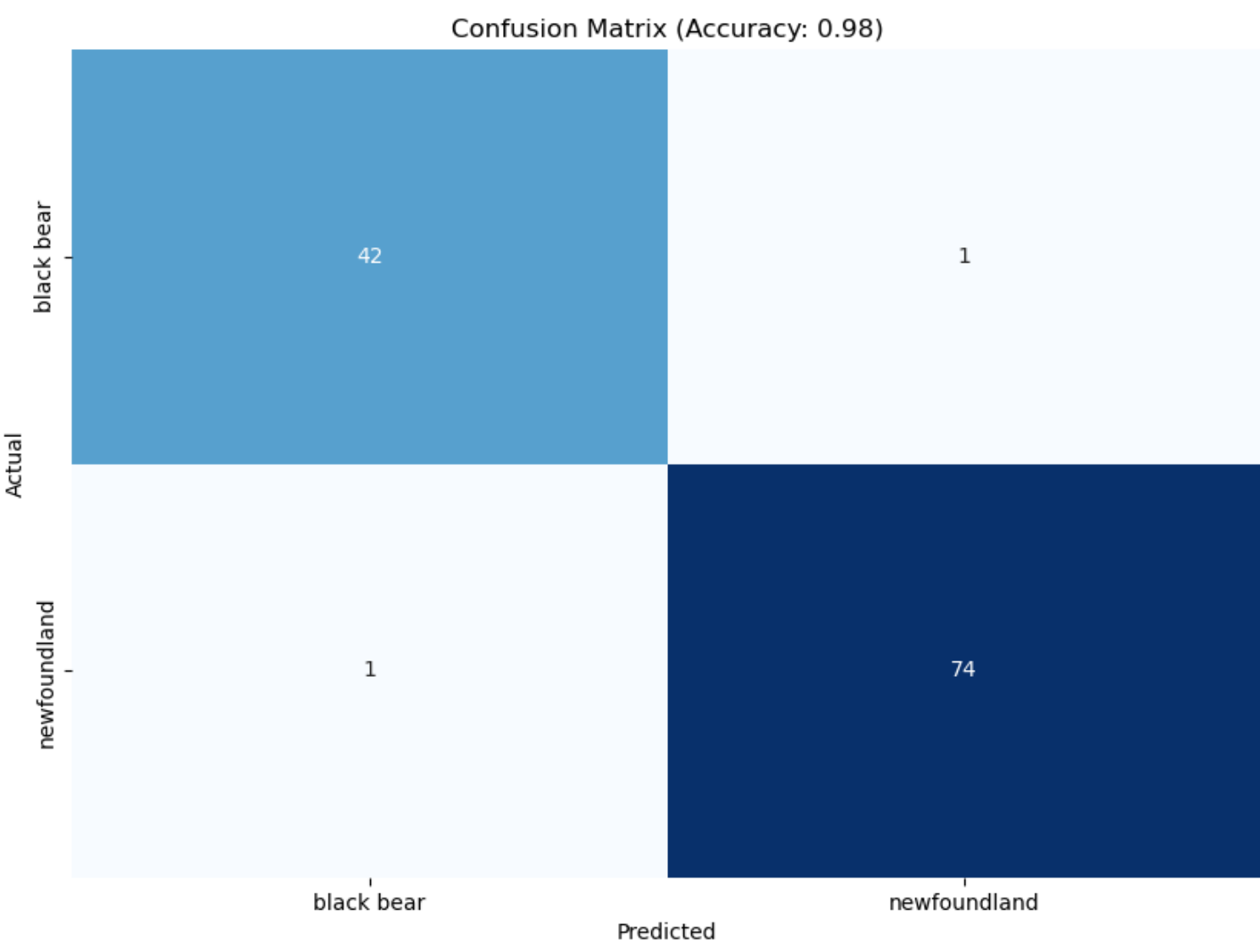
Confusion Matrix

True Positives (Black Bear): 42

False Positives: 1

True Negatives (Newfoundland): 74

False Negatives: 1



Classification Report

The model achieved high performance metrics with targets of 0.98 for black bears and 0.99 for Newfoundland dogs.

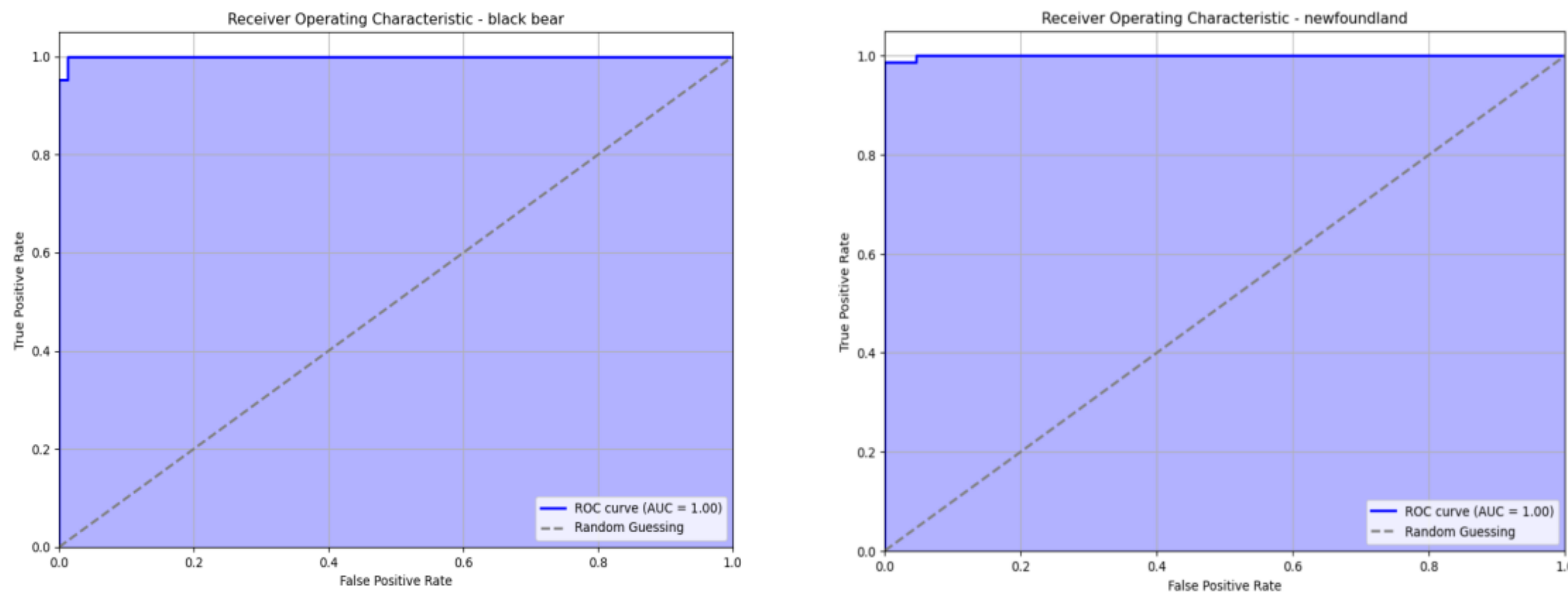
	precision	recall	f1-score	support
black bear	0.98	0.98	0.98	43
newfoundland	0.99	0.99	0.99	75

The overall accuracy of the model is 0.98, demonstrating its effectiveness in distinguishing between the two classes.

accuracy			0.98	118
macro avg	0.98	0.98	0.98	118
weighted avg	0.98	0.98	0.98	118

ROC Curve

The ROC curve demonstrates the model's ability to distinguish between classes, with an area under the curve (AUC) of 1.00, indicating perfect performance.



UNSEEN DATASET PREDICTION



FUTURE WORK

- Explore methods to combine different neural network architectures, such as integrating Convolutional Neural Networks (CNNs) with Recurrent Neural Networks (RNNs), to effectively process video data or time series data.
- Train the model to recognize a greater variety of animal species, enhancing its adaptability in different environments and conditions.
- Incorporate the model into real-time systems, such as mobile applications or drones, for instant on-site species identification.

REFERENCES

Datasets: <https://www.kaggle.com/datasets/hoturam/bear-dataset>
<https://www.kaggle.com/datasets/imbikramsaha/dog-breeds>

Muhammad Abdullah. (2024). Hotdog or Not: The Ultimate Classifier.
<https://www.kaggle.com/code/abdmmental01/hotdog-or-not-the-ultimate-classifier>